

Weather App Project - Detailed Learning Notes

Fetching Weather Data

Concept: Using `fetch()` to call the World Weather Online API.

Code:

```
const response = await fetch(`${url}&q;=${place}#_of_days=1`);  
const data = await response.json();
```

Explanation: `fetch()` sends a request to the API. `response.json()` converts the returned JSON text into a JavaScript object.

Current Time

Code:

```
const currentTime = new Date().toLocaleTimeString([], {hour:'2-digit', minute:'2-digit'});
```

Explanation: Creates a `Date` object and formats the current time without seconds.

Current Day and Date

Code:

```
const day = new Date().toLocaleDateString('en-US',{weekday:'long'});  
const date = new Date().toLocaleDateString('en-IN',{day:'numeric',month:'long',year:'numeric'});
```

Explanation: Generates readable day and date strings.

Capitalizing City Names

Code:

```
function capitalize(str){ return str.charAt(0).toUpperCase()+str.slice(1); }
```

Explanation: Converts the first character to uppercase while keeping the rest unchanged.

DOM Selection Issues

Problem: Cannot set properties of null.

Fix:

```
document.querySelector('#place').innerText = place;
```

Explanation: The selected element must exist in the HTML before updating it.

Weather Icons

Code:

```
document.querySelector('.text img').src = data.data.current_condition[0].weatherIconUrl[0].value;
```

Explanation: Image URLs returned by the API can be assigned directly to an `img` element's `src`.

Creating Elements Dynamically

Code:

```
const card = document.createElement('div');
card.classList.add('hour-card');
container.appendChild(card);
```

Explanation: Generates forecast cards using JavaScript instead of hardcoding HTML.

Hourly Forecast Cards

Code:

```
hourly.forEach(hour => { ... });
```

Explanation: Loops through hourly weather data and creates a card for each time slot.

Horizontal Scrolling

CSS:

```
.today-weather { display:flex; overflow-x:auto; }
.hour-card { flex-shrink:0; min-width:120px; }
```

Explanation: Prevents cards from shrinking and allows sideways scrolling.

Hiding Scrollbars

CSS:

```
scrollbar-width:none;
.today-weather::-webkit-scrollbar{display:none;}
```

Explanation: Keeps scrolling functionality while hiding the scrollbar.

City Search

Code:

```
document.querySelector('#searchBtn').addEventListener('click', () => { getWeather(cityInput.value);
});
```

Explanation: Reads user input and fetches weather data for the chosen city.

GitHub and API Keys

Important: Do not commit active API keys to public repositories.

Replace:

```
const key = 'REAL_KEY';
```

with:

```
const key = 'YOUR_API_KEY';
```

Explanation: Prevents others from using your quota.